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From Harvest to Food Security: Machine Learning Forecasts for the United States

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Abstract

Food insecurity persists even in high-output agricultural systems, underscoring the need for timely national indicators. This study tests whether U.S. agricultural production signals can predict a composite Food Security Index (FSI) using an annual dataset (1970-2023) integrating USDA NASS QuickStats measures for three major staple crops: corn, soybeans, and wheat (yield, production, area planted, and area harvested).

The prediction target is an FSI scaled $[0, 1]$ (higher = more food secure) constructed from four FAOSTAT variables: dietary energy supply, protein supply, GDP per capita, and a food production index. We compared multiple supervised regression models with training on 1970-2012 and testing on 2013-2023 data. We further included a year to year change (ΔFSI), with engineered shock & lag features. A Random Forest model trained on the ΔFSI feature set performed the best, testing Root Mean Squared Error (RMSE) ≈ 0.066, and improving on a persistence baseline.

Overall, results showed that the national crop statistics contain measurable predictive indicators for annual food security dynamics and motive extensions.

Keywords:
Agriculture, Food Security Index, Machine Learning

1 Introduction

Food insecurity persists even in highly productive agricultural systems because national food security depends on more than total crop output [1]. It also depends on whether people can reliably access safe, sufficient, nutritious food and whether the system remains resilient to shocks [2]. With climate variability [3], market disruptions [4], and economic pressures increasingly interacting, there is growing demand for timely, interpretable national indicators that can be tracked consistently over long periods to support decision making [5].

Agricultural production is among the most consistently measured national variables. In the United States, corn, soybeans, and wheat are especially well documented because they dominate staple crop production and shape downstream food and feed markets [6, 7]. Annual changes in yields, acreage, and output can reflect both biophysical constraints and land-use responses. However, production alone does not map cleanly onto national food security outcomes: access and affordability [8], supply chains [9, 10], and broader macro conditions introduce nonlinearities and time-lagged effects.

Researchers have developed multiple food-security indicators to reflect food security's multifaceted nature. The Global Food Security Index (GFSI) [11, 12], for example, assesses 113 countries using measures of affordability, availability, quality and safety, and sustainability/adaptation, offering a defensible, simple, and transparent basis for agricultural policymaking [13, 14].

In this study, a composite Food Security Index (FSI) is developped for US agricultural system using four Food and Agriculture Organization of the UN's corporate statistical database (FAO-

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STAT) indicators [15]: dietary energy supply, protein supply, GDP per capita, and the food production index. This macro summary score tracks year-to-year national food security with long-run consistent data and serves as a proxy for overall conditions (not household hunger). Energy supply represents availability [16], protein supply reflects dietary adequacy [17], GDP per capita captures economic access [18], and the production index indicates domestic supply relative to a base period [19].

2 Materials & Methods

2.1 Data and Study Design

We assessed whether U.S. staple-crop production predicts a composite food security index (FSI) using an annual time series from 1970-2023 [20, 21]. Crop predictors were drawn from United States Department of Agriculture (USDA) National Agricultural Statistics Service (NASS) QuickStats for corn, soybeans, and wheat, including yield, total production, and harvested area for each crop (N=1,944) [22]. The response variable was an FSI scaled to $[0, 1]$, with higher values indicating better food security.

In addition to crop exclusive statistics, we included exogenous drivers: a food Consumer Price Index (CPI) and its annual change, World Bank "Pink Sheet" indices (Energy, Agriculture, Food, Grains, Fertilizers, Metals & Minerals) and selected commodity prices (crude oil, maize, soybeans, and wheat—Hard Red Winter and Soft Red Winter) [15]. All datasets were restricted to the U.S., aggregated to annual values, and merged by year into a master table; non-numeric entries were treated as missing and dropped only when required for model fitting to preserve predictor-target alignment. The resulting dataset was then used for feature engineering (lags and year-over-year "shock" variables) and model evaluation using time-ordered train/test splits.

Figure 1 summarizes the end-to-end flow (data integration through validation). To reduce bias and prevent leakage, we excluded state-level QuickStats features, ensuring model development and evaluation reflected the national U.S. context.

2.2 Data Preparation

Crop predictors were retrieved via the USDA NASS QuickStats API [22] by querying U.S.-level annual records for corn, soybeans, and wheat and extracting yield, production, and harvested area. API calls were executed in Python using the `requests` library, and each metric was saved as a structured CSV. Exogenous series were obtained from the U.S. Bureau of Labor Statistics (food CPI) [23], the World Bank Commodity Price Data ("Pink Sheet") for indices and commodity prices [? 24, 25], and World Bank population totals.

All preprocessing was performed in Python [26] using `pandas` [27] and `NumPy` [28] for parsing, numeric conversion, reshaping, merging, and feature engineering (lag and shock variables). Non-numeric entries were treated as missing and removed only when required for model fitting; `pathlib` supported file handling [29], and scikit-learn utilities supported model-ready preprocessing [30]. Cleaned series were aggregated to one row per year and merged with the constructed FSI and exogenous predictors to form the final 1970-2023 modeling dataset. Continuous features were standardized using scikit-learn's `StandardScaler` to improve comparability across units and optimize model performance.

2.3 Targets and Feature Engineering

We created the FSI in multiple steps. For $FSI(t)$, we defined it as a single number between 0 and 1 for each year t, where the higher the value, the better food security (lower insecurity). We built it off of four annual national indicators:

1. K(t): Dietary energy supply (kcal per capita per day)
2. P(t): Protein supply (g per capita per day)
3. G(t): GDP per capita (USD)
4. F(t): Food production index

Because these four variables have different units and ranges, we first normalized each one to the same [0, 1] scale as the final FSI over our full study cohort of 1970-2023, making them directly comparable. By using min-max normalization:

$$X_{\text{norm}}(t) = \frac{X(t) - \min(X)}{\max(X) - \min(X)}$$

for

$$X \in \{K, P, G, F\}$$

Each component lies in a range of [0,1] after this.

Secondly, we combined all four components into one index by equal-weighted arithmetic mean. This is because we did not set any weights

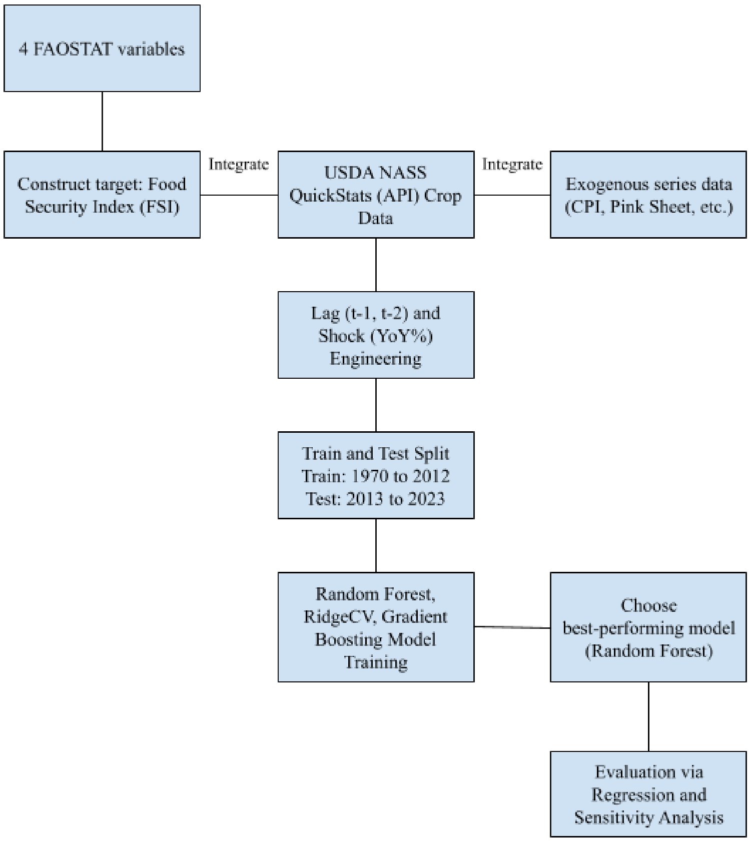


Figure 1: Study Flow Chart

to any of the components, making all of them of equal importance to one another.

$$FSI(t) = \frac{K_{\text{norm}}(t) + P_{\text{norm}}(t) + G_{\text{norm}}(t) + F_{\text{norm}}(t)}{4}$$

This, in turn, makes $FSI(t)$ a value within [0,1], where a higher FSI value corresponds to stronger food security conditions, by our composite definition. At the same time, a drop in FSI would signify a deterioration driven by one or more components declining in relevance to their historical range.

While FSI provides an interpretable annual picture, its strong persistence over time reflects the idea that forecasting will be easier in year-to-year change increments rather than absolute levels. Therefore, we tested two prediction formulations: level and change-based. In the level formulation, models predict $FSI(t)$ directly, where $FSI(t)$ denotes the FSI in year t. In the change based formulation, models predict annual change $\Delta FSI(t) = FSI(t) - FSI(t-1)$ and reconstruct levels via $FSI'(t) = FSI(t-1) + \Delta FSI'(t)$. Where this year's predicted FSI equals last year's FSI plus the model's predicted change this year. The FSI' means that value is a predicted FSI. Predictors included crop indicators (yield, production, and area harvested for corn, soybeans, and wheat) as well as exogenous variables in a climate and economic context.

To capture disruptions and delayed effects, we manually created shock features and lags for all time-varying predictors. Shocks were computed on a year by year basis as changes .(percentage changes for strictly positive variables) We also included 1 to 2 year lags of both raw and shocked level terms. This design allows our model to learn immediate versus delayed associations between production, external conditions, and changes in national food security.

2.4 Model Selection

To accurately test whether or not all of the predictors could contain a usable signal for food security, we employed multiple supervised machine learning regression models to predict for our FSI. The model had a time-respective training and testing data split where 1970 to 2012 data was used for training and 2013 to 2023 data was used for testing. The model was benchmarked against a persistent baseline that forecasts the next year's value.

We compared a small set of regression models to test, each with different strengths. These included a regularized Ridge regression linear model as well as a non linear Random Forest regression ensemble model. The Ridge model provides a transparent benchmark that handles all correlated predictors through regularization; the Random Forest model can help capture non-linear interactions between variables without requiring a separate fixed functional form. To evaluate which model allowed for easier forecasting, our two formulation processes were used, and the best-performing approach was selected based on predictive accuracy and the held-on test period.

2.5 Performance Metrics

Predictive accuracy on held-out years was measured using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2 values, reported in relevance to the persistence baseline. For the second formulation, the change-based formulation, we assessed directional performance additionally by classifying whether ΔFSI was bigger than 0 and then computing a confusion matrix and balanced accuracy along with it. Receiver Operating Characteristic Area Under the Curve (ROC AUC) and average precision were also calculated using the changed, predicted scores.

3 Results

Through national staple-crop statistics and exogenous economic indicators, we evaluated the predictive perforation of the proposed Ridge and Random Forest models for estimating food security in the United States. In this section, we see how well our models predict U.S. food security in the held-out period and what signals drive said predictions.

According to the results across the tested models, the Random Forest model, trained on ΔFSI, was selected as the primary approach as it provided the most reliable out-of-sample tracking of FSI relative to the alternative models and our persistence baseline. Therefore, we present the results from the Random Forest model first, followed by diagnostics and an auxiliary directional evaluation.

Figure 2 shows a comparison between the observed FSI in the 2013 to 2023 test period and the predictions from the Random Forest model trained on annual changes (ΔFSI). In addition, a persistence baseline that simply carries forward the previous FSI ($FSI(t-1)$) is also compared. We set the y-axis to start at 0.60, rather than 0, to focus on the 2013-2023 range and make small but meaningful differences between the observed FSI, model predictions, and the persistence baseline easier to see. Overall, the Random Forest model is able to forecast patterns following the same broad trajectory as the observed series, including a mid-decade decline and substantial rise right after 2017. This suggests that the model captures meaningful year by year movements rather than only the trend in the long run. The persistence baseline also helps track the general direction; however, it responds more slowly to turning points because it assumes that there has been no change beyond last year's level. Differences between predicted and observed curves are most visible around the sharper transitions, where the model either slightly over or underestimates the magnitude of change. But, the close alignment across the vast majority of the years shows that crop and "shock and lag" features contain usable signals for national food-security dynamics.

Figure 3 plots the same test-year values as Figure 2, but collapses them into point pairs where each dot is one year with the x-axis being the observed FSI and the y-axis as the predicted FSI. Points closer to the line of best fit means better agreement. In our test set, our predictions are aligned with the observations with a correlation value of around 0.82, with 6 out of 11 years falling within 0.05 of the true value and 4 out of 11 within 0.02. Similarly to Figure 1, the largest mismatches occur in the volatile years: 2017 being one as it is underpredicted.

Figure 2 is a time-series story which shows whether the model follows a yearly trajectory with turning points compared to a persistence baseline. On the other hand, Figure 3 is a calibration, or fit, check as it ignores ordering in time and visualizes how close predictions in our model are to the correct values overall.

Figure 4 shows the residual of the model, an annual year-by-year prediction error in the testing period. This is defined by the equation: $r(t) = FSI_{\text{actual}}(t) - FSI_{\text{pred}}(t)$. Positive results indicate underprediction, where the observed FSI was higher than the estimation made by the model, while negative residuals indicate overpredication. The centre line at Residual $= 0$ means that there is "no error" where the model predicted the FSI exactly for that year. Across 2013 to 2023, residuals are generally small and only fluctuate around zero, where the mean residual is about $+0.016$ and the median is around -0.016. This suggests that the model is not consistently biased in one signal direction over the entire testing window.

However, the plot also shows the years that drive most of the error. The largest underpredic-

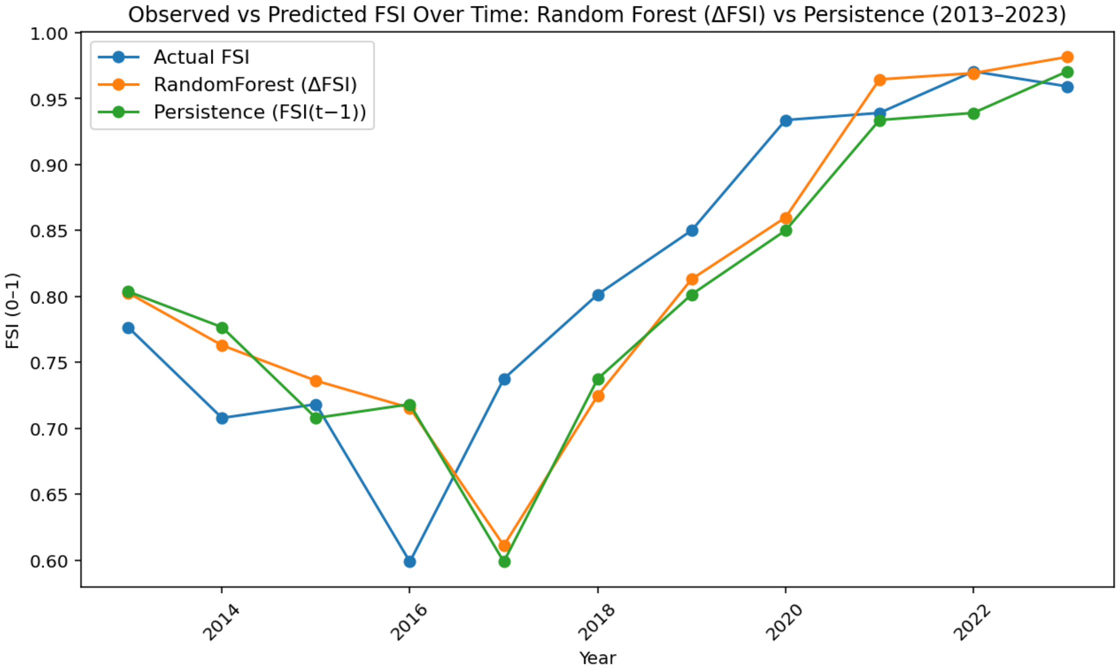


Figure 2: Observed vs Predicted ΔFSI Over Time: Random Forest vs Persistence (2013-2023)

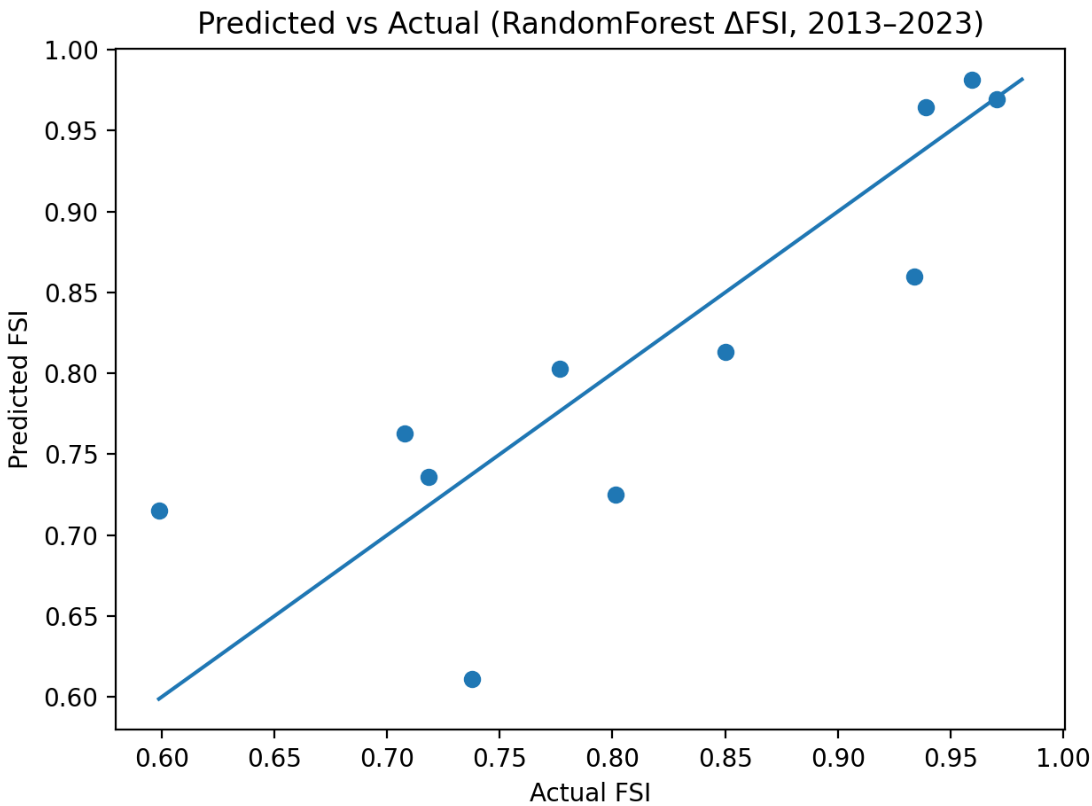


Figure 3: Predicted vs Observed FSI in the Test Period (2013-2023)

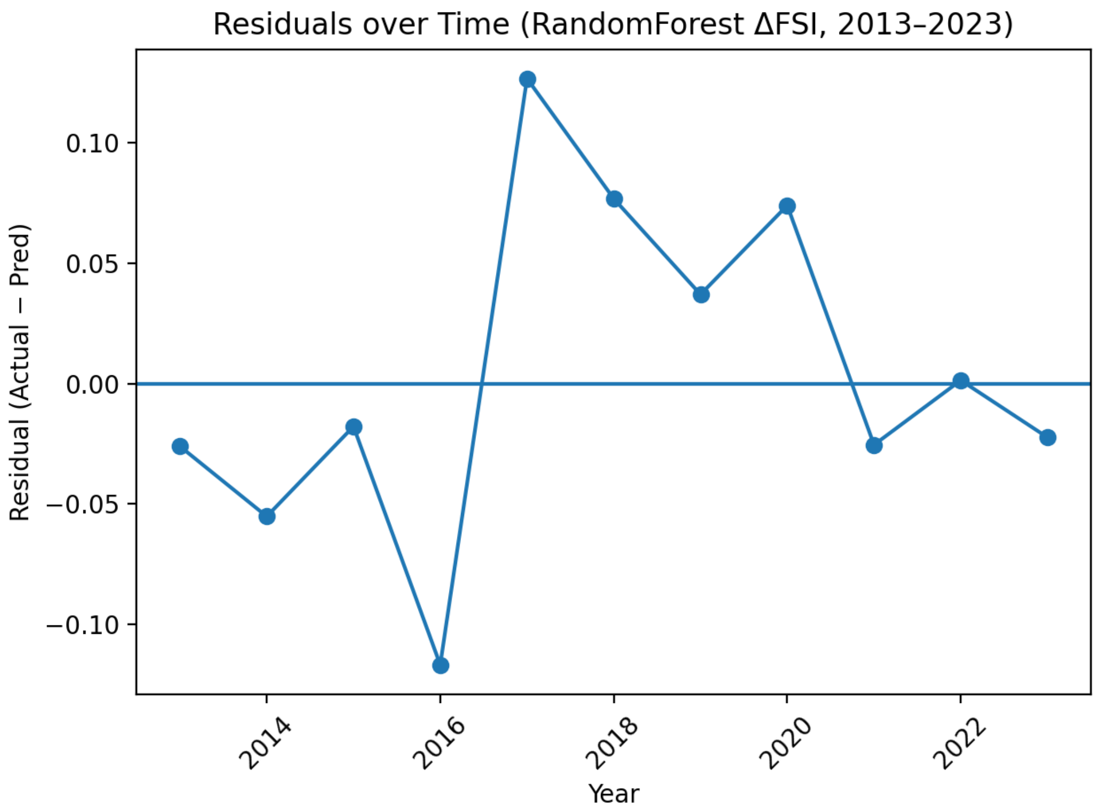


Figure 4: Residuals Over Time (Observed - Predicted FSI), 2013-2023

tion occurs in 2017, where the residual is $+0.150$, with an actual FSI value of 0.738 and a predicted value of 0.588. The largest overprediction happens in 2016, with a residual of -0.108 and a split of 0.599 as the actual values versus the predicted 0.707. These spikes cluster in the sharp transitions in the FSI trajectory, further showing that the model struggles to adapt when food security shifts abruptly. Overall, the residual pattern supports the interpretation that our model is able to track gradual movement well but extreme changes still remain the biggest hardships.

The importance of features for predicting FSI was evaluated based on the model coefficients. Features with larger absolute coefficient values were deemed more significant, indicating a stronger influence on the target outcome.

Figure 5 shows which predictors the Random Forest model relied on the most when forecasting ΔFSI. This importance ranking is dominated by production-side crop variables and short-term temporal structure, especially on the wheat and corn warians. The most influential predictor was "wheat harvested area (t-1)" with an importance of around 0.064, followed closely by "corn yield (level)" at around 0.060. Other high ranking features are also lagged crop quantities such as "wheat product (t–2)" (≈ 0.040), "wheat yield (t-2)" (≈ 0.036), and "wheat production (t-1)" (≈ 0.030). This indicates that the model draws heavily on one to two year delayed information rather than contemporaneous levels. This means that lag features (labels such as t-1 and t-2) are simply predictor values that are taken from previous years, allowing the model to capture delayed effects where impacts on food security unfold over time.

In addition, shock-style variables also appear repeatedly. Features labeled with "(YoY %)" are a representation of year by year percent change, meaning how much a variable has increased or decreased compared to the previous year. In Figure 4, these YoY % are the shock variables to help the model distinguish between short-run departures and typical conditions. The appearance of "soybeans yield (YoY %)" (≈ 0.033), "soybeans production (YoY %)" (≈ 0.029), and "corn yield (YoY %)" (≈ 0.026) among the top predictors suggests that year by year sudden changes do contain useful signal for changes in national food security.

Finally, market price factors also contribute but do not dominate: "U.S. wheat price (SRW) (YoY %)" (≈ 0.024), "maize price (YoY %)" (≈ 0.019), and "U.S. wheat price (HRW)" (YoY

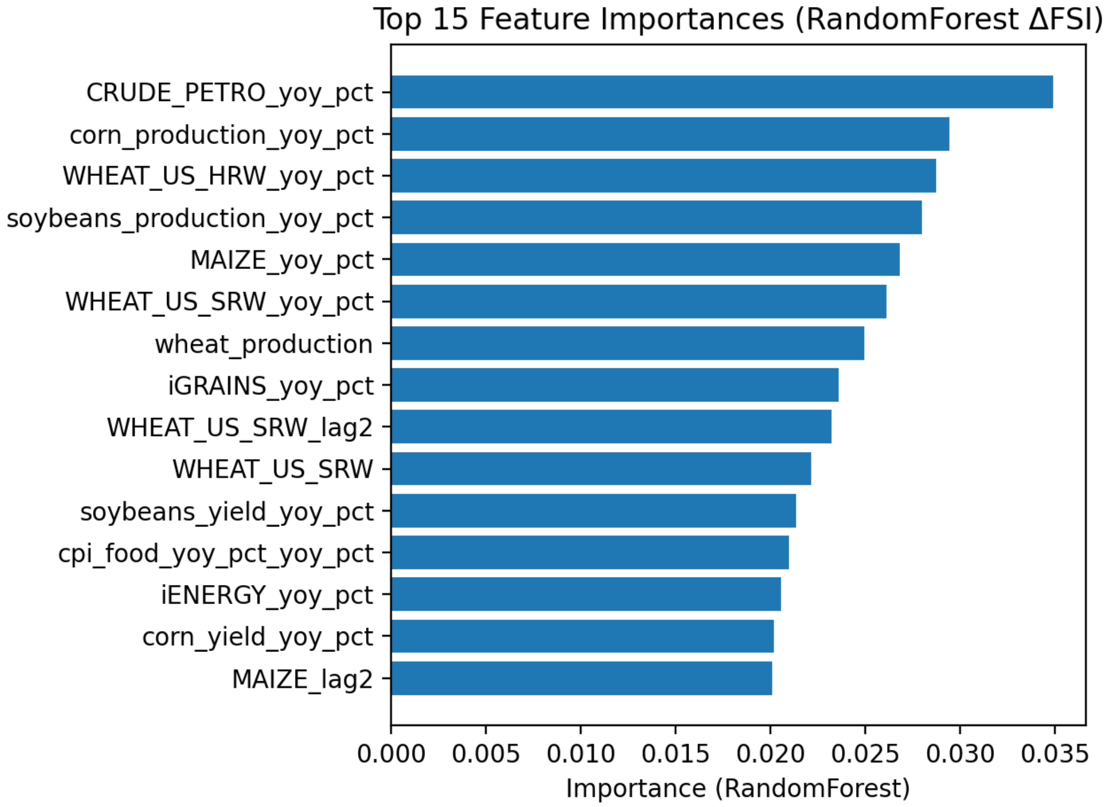


Figure 5: Top 15 Feature Importances: Random Forest ΔFSI

%) ($\approx$ 0.016) are included in the top 15, which implies that these commodity-price shocks help provide an additional contest beyond production volumes. Most importantly, these reflect predictive contributions rather than just casual effects. Figure 5 indicates which variables were most useful in reducing prediction error when the Random Forest learned patterns ΔFSI.

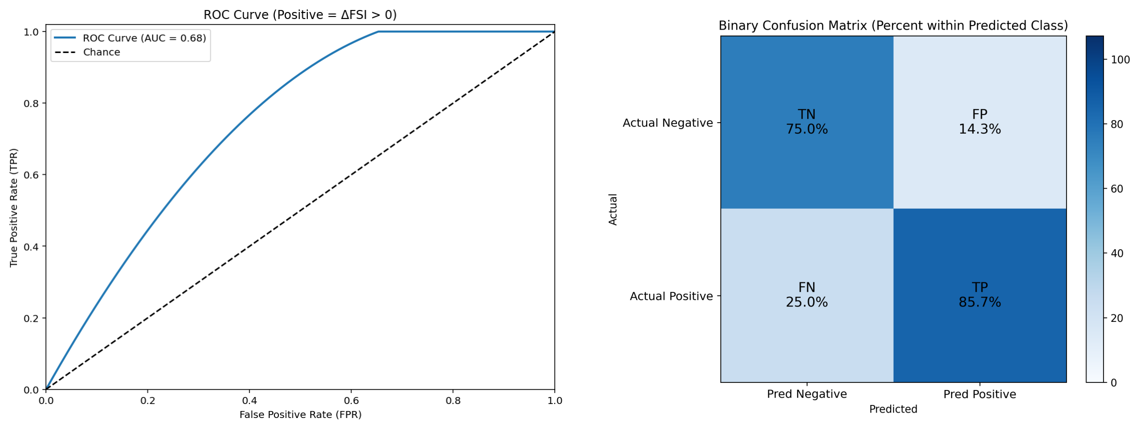
Table 1 shows how sensitive the Random Forest ΔFSI model is to the way we encoded short-run disruptions and delayed effects in our predictors. We varied two feature types: our shock features (year by year percentage changes, "_yoy_pct") and our lag features (prior year values, "_lag1" or "_lag2"). This setup helps isolate whether the model benefits more from immediate year by year changes, historical memory, or both.

The pattern is evident: accuracy suffers when temporal structure is removed, but accuracy is consistently improved when shocks are combined with short lags. Error increases significantly (RMSE 0.0788, $\mathbf{R^2}$ 0.5478) when both shocks and lags are eliminated ("no shocks + no lags"), and the use of only shocks or only lags stays weak (RMSE 0.0791 and 0.0809, respectively). On the other hand, when shocks and lags are combined, error is significantly reduced and performance approaches the persistence benchmark (RMSE 0.0699). Shocks + lag1+lag2 (no lagged shocks) is the grid's best-performing configuration (RMSE 0.0701, MAE 0.0509, $\mathbf{R^2}$ 0.6419), and lag2 shock setups also enhance directional performance (balanced accuracy 0.7143). Overall, Table 1 demonstrates that the model best captures the dynamics of national food security when it can learn from both YoY shocks and 1-2 year delayed effects rather than just static levels alone.

Figure 6 is an evaluation on whether the model is able to predict the direction of food security change on an annual scale by framing the outcome as a binary label. Our definitions included: Positive if $\Delta FSI > 0$, if FSI improved from the previous year, and Negative if ΔFSI is $<= 0$. This figure includes two visualizations. The ROC curve (6a) helps summarize how well the model is able to distinguish between these two classes across all possible decision thresholds. With an AUC of 0.68, it indicated modest to moderate discrimination, which is better than chance (0.50), but also not the most strong. Our second visualization (6b) is the confusion matrix which is shown as percentages with the predicted classes (normalized by column), so each column is a reflection of how accurate the model's pred-

Table 1: Shock-Lag Sensitivity Analysis for the Random Forest ΔFSI Model (Test: 2013-2023)

Configuration	**Shocks**	**Lag Depth**	**Lagged Shocks**	**RMSE**	**MAE**	**R^2**	**Balanced Accuracy**
Persistence baseline (FSI(t)=FSI(t-1))	—	—	—	0.0699	0.0554	0.6443	
Shocks + lag1+lag2 (no lagged shocks)	On	2	Off	0.0701	0.0509	0.6419	0.7143
Shocks + lag1+lag2	On	2	On	0.0704	0.0497	0.6393	0.7143
Shocks + lag1	On	1	On	0.0712	0.0517	0.6308	0.6429
No shocks + no lags	Off	0	Off	0.0788	0.0646	0.5478	0.5
Shocks only	On	0	Off	0.0791	0.0592	0.5438	0.6429
Lag1 only	Off	1	Off	0.0809	0.071	0.5227	0.5



(a) ROC curve

(b) Confusion Matrix

Figure 6: ROC Curve and Confusion Matrix for Predicting $\Delta FSI > 0$

icated label is. When the model predicts a Negative year, 75.0% of those predictions are truly negative (TN) and 25.0% are actually positive (FN). When the model predicts a Positive year, 85.7% are truly positive (TP) and 14.3% are false positives (FP).

Table 2 is a comparison of directional performance across a full set of models by evaluating whether each approach correctly predicts the sign of annual food security change, where Positive is $\Delta FSI > 0$ and Negative is $<= 0$. While the earlier results focus exclusively on the Random Forest ΔFSI model for detailed diagnostics, Table 2 is a showcase of the other models we tried to test with by contextualizing how alternative methods and feature sets perform on the same 2013 to 2023 test window.

The strongest directional performance was achieved by the Random Forest model using our fully engineered feature set (ΔFSI: crops + macro + shocks + lag), which achieved a balanced accuracy of 0.8036, with sensitivity 0.8571 and specificity 0.7500. With ΔFSI: macro + shocks + lags, the Random Forest model achieved a balanced accuracy of 0.7321, with sensitivity 0.7143 and specificity 0.7500. For ΔFSI: crops + shocks + lags, Gradient Boosting reaches a balanced accuracy of 0.6786 (sensitivity 0.8571, specificity 0.5000), suggesting it is comparatively good at catching improvement years but less consistent at correctly identifying non-improvement years. Under the same reduced data set, RidgeCV yields a balanced accuracy of 0.6250 with perfect sensitivity (1.0000) but low specificity (0.2500).

4 Discussion

Overall, our results showed that national crop yield, production, and harvest area statistics helps signal tracking for U.S. food security over time. The Random Forest ΔFSI model consistently provided the most reliable results out of all the tested approaches. We discovered that supervised models can learn relationships

Table 2: Directional Model Comparison for ΔFSI:
Balanced Accuracy, Sensitivity, and Specificity (Test: 2013-2023)

Setup	**Model**	**Balanced Accuracy**	**Sensitivity**	**Specificity**
ΔFSI: crops+macro+shocks+lags	RandomForest	0.8036	0.8571	0.75
ΔFSI: macro+shocks+lags	RandomForest	0.7321	0.7143	0.75
ΔFSI: crops+shocks+lags	GradientBoosting	0.6786	0.8571	0.50
ΔFSI: crops+shocks+lags	RidgeCV	0.625	1	0.25
Baseline	Persistence ($\Delta = 0$)	0.5	0	1

between staple-crop indicators and a composite FSI using an annual U.S.-only dataset (1970 to 2023), particularly when the target is modeled as year-to-year change (ΔFSI) rather than absolute levels. The Random Forest ΔFSI framework performed competitively against a strong persistence benchmark and generally followed the observed trajectory during the 2013 to 2023 holdout period. This lends credence to the notion that national crop conditions frequently correspond with more general food-system outcomes, making them useful for forecasting and monitoring.

A major takeaway is that our shock and lag feature engineering mattered as much as model choice. The shock-lag sensitivity results show that removing both YoY shock terms and lag structure noticeably worsened performance, whereas including both consistently improved results.This is plausible in food systems where production outcomes, input costs, and price conditions influence food availability and stability with delays rather than instantaneously. In addition, lagged crop quantities ("t-1" and "t-2") and shock terms ("YoY %") were amongst some of the strongest predictors in Figure 4, implying that the model benefits from both the "memory" of prior conditions as well as a detection of sudden year by year disruptions.

At the same time, our results clarify what exclusively crop-based prediction can and cannot do. The residual analysis shows that the largest errors occur around years with sharper transitions, and the directional evaluation shows that for up or down changes in ΔFSI, there is only moderate discrimination with an AUC of 0.68. This is consistent with the broader reality of food security as it is not solely determined by production; affordability, distribution, macroeconomic shocks, and policy/trade dynamics all have potential to shift outcomes even if harvest data shows strength. As a result, production and commodity indicators are most useful for identifying medium-term trends and gradual movement, and less trustworthy when changes in access or policy predominate in the food security signal.

Since directional classification is more difficult than level prediction in this context, the AUC is not higher. First, a few borderline years can significantly alter the ROC curve and lower AUC because the test window only spans 11 years. Second, the positive/negative boundary is susceptible to small noise in either the predictors or the constructed FSI itself because many year-to-year changes in a national index are small (near ΔFSI=0). Third, crop and commodity indicators may track the overall movement of FSI without consistently capturing the precise up/down. This is because, even when agricultural production contains a signal, the sign of food-security change yearly also depends on drivers that are only partially represented in our features, such as policy shifts, trade dynamics, household prices, distribution, and access constraints. For these reasons, our ROC/AUC results are a supporting diagnostic where the model shows some ability to detect improvement years.

However, when interpreting these results, a number of limitations should be taken into account. Due to the short test window from 2013 to 2023, performance metrics are sensitive to a few borderline and inconsistent years. Localized shocks that do not significantly register in aggregates may be missed by the national annual design, which smoothes within-country variation. Although the FSI offers a consistent 0 to 1 target, any index depends on design decisions that may impact year-to-year dynamics. The FSI itself is a constructed composite based on modeled transformations of FAOSTAT variables. Lastly, despite the fact that we included a number of macro and commodity-price series, the feature set leaves out important factors that affect food security, such as measures of household income and poverty, changes in policy and assistance, trade volumes, inventories and stock-to-use conditions, and distribution restrictions.

Otherwise, our findings have a direct relevance to policy. Firstly, it shows the use of routinely and easily accessible crop statics as a part of an early-warning style monitoration. Our approach is best-suited to flagging when the U.S. system is deviating from recent norms as we use the best-forming setups including YoY shocks and short lacks. This can help agencies and decision-makers prioritize attention, scenario planning, and preparedness. Secondly, our limitations are also equally informative. Moderate ROC/AUC performance and outlier years, like in 2017, reinforce the idea that supply-exclusive improvements alone does not guarantee better food security. Implying that yield-boosting or acreage-expansion policies should be paired with measures that target affordability and access, such as food price stabilization, safety net strengthening, or improving distribution reliability, our limitations actually help and inform those in power of the main goal of translating production strength into real gains in food-security.

In conclusion, crop indicators by themselves cannot fully explain food security, a multi-layer commodity, but they are sufficiently informative to support tracking and monitoring at the national level, particularly when modelled with shocks and lags. The most compelling evidence from our work is that expanding the predictor set toward access, trade, and policy drivers is the most promising route to policy-relevant forecasting, and that year by year changes in food security are more predictable when models can learn from both short-run disruptions and delayed effects.

5 Conclusion

Even in a high-producing agricultural system, such as the United States, national food security can also shift, which makes timely monitoring and early warnings especially important. In conclusion, crop indicators by themselves cannot fully explain food security, but they are sufficiently informative to support tracking and monitoring at the national level, particularly when modelled with shocks and lags. The most compelling evidence from this work is that expanding the predictor set toward access, trade, and policy drivers is the most promising route to policy-relevant forecasting, and that year to year changes in food security are more predictable when models can learn from both short-run disruptions and delayed effects. Future work should focus on further increasing our predictor set to include more variables such as access, trade, and policy drivers and move toward uncertainty-aware forecasts. That allows for decision-makers to more effectively allocate resources and respond earlier to conditions that threaten national food security. This approach has the potential to respond to noticed issues faster and more effectively identifying food insecurity risk.

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